**Example 7.5**

Find the sum $A + B + C$ if $A, B, C$ are given by

$$A=\begin{bmatrix}\sin^2\theta & 1\\ \cot^2\theta & 0\end{bmatrix},\ B=\begin{bmatrix}\cos^2\theta & 0\\ -\operatorname{cosec}^2\theta & 1\end{bmatrix} \text{ and } C=\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}.$$

**Solution**

By the definition of sum of matrices, we have

$$A+B+C=\begin{bmatrix}\sin^2\theta+\cos^2\theta+0 & 1+0-1\\ \cot^2\theta-\operatorname{cosec}^2\theta-1 & 0+1+0\end{bmatrix}=\begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix}.$$

**Example 7.6**

Determine $3B + 4C - D$ if $B, C,$ and $D$ are given by

$$B=\begin{bmatrix}2 & 3 & 0\\ 1 & -1 & 5\end{bmatrix},\ C=\begin{bmatrix}-1 & -2 & 3\\ -1 & 0 & 2\end{bmatrix},\ D=\begin{bmatrix}0 & 4 & -1\\ 5 & 6 & -5\end{bmatrix}.$$

**Solution**

$$3B+4C-D=\begin{bmatrix}6 & 9 & 0\\ 3 & -3 & 15\end{bmatrix}+\begin{bmatrix}-4 & -8 & 12\\ -4 & 0 & 8\end{bmatrix}+\begin{bmatrix}0 & -4 & 1\\ -5 & -6 & 5\end{bmatrix}=\begin{bmatrix}2 & -3 & 13\\ -6 & -9 & 28\end{bmatrix}.$$

**Example 7.7**

Simplify :

$$\sec\theta\begin{bmatrix}\sec\theta & \tan\theta\\ \tan\theta & \sec\theta\end{bmatrix}-\tan\theta\begin{bmatrix}\tan\theta & \sec\theta\\ \sec\theta & \tan\theta\end{bmatrix}.$$

**Solution**

If we denote the given expression by $A$, then using the scalar multiplication rule, we get

$$A=\begin{bmatrix}\sec^2\theta & \sec\theta\tan\theta\\ \sec\theta\tan\theta & \sec^2\theta\end{bmatrix}-\begin{bmatrix}\tan^2\theta & \tan\theta\sec\theta\\ \sec\theta\tan\theta & \tan^2\theta\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

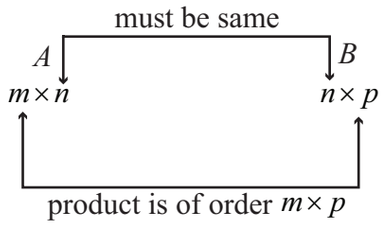
## (3) Multiplication of matrices

**Definition 7.15**

A matrix $A$ is said to be **conformable for multiplication** with a matrix $B$ if the number of columns of $A$ is equal to the number of rows of $B$.

That is, if $A=[a_{ij}]_{m\times n}$ and $B=[b_{ij}]_{n\times p}$ are given two matrices, then the product of matrices $A$ and $B$ is denoted by $AB$ and its order is $m \times p$.

The order of $AB$ is $m\times p=(\text{number of rows of } A)\times(\text{number of columns of } B)$.






07_11th-Maths-Vol2-Unit7.indd 9 16-12-2022 10:59:38